

C975

Science Methods—Middle Grades General Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher whose classroom just got a new interactive projector writes an objective: “Students will analyze temperature data they collected to find patterns in how sunlight heats land and water.” Which dimension is “analyze data”?

- A. A crosscutting concept
- B. A disciplinary core idea
- C. A science and engineering practice
- D. Not a dimension; data analysis is only a math skill

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In the objective “Students will analyze temperature data they collected to find patterns in how sunlight heats land and water,” which part is the crosscutting concept?

- A. Patterns
- B. Analyzing data
- C. How sunlight heats land and water
- D. The thermometers used to collect data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A team at a school that runs a Saturday science fair plans a unit toward a performance expectation: week 1, vocabulary for the core idea; week 2, graphing practice with unrelated data; week 3, a lecture on patterns. What is the main problem?

- A. It needs more vocabulary before any practice.
- B. The dimensions are taught separately instead of used together to make sense of the core idea.
- C. Nothing; order never matters because performance expectations are assessed at the end.
- D. Three weeks is too short.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At a school whose lab was renovated last summer, a teacher is choosing an anchoring phenomenon for a unit on chemical reactions. Which is strongest?

- A. "Chemical reactions"
- B. A tour of the renovated lab
- C. The definitions of reactant and product
- D. A sealed bag of baking soda and vinegar puffs up and feels cold to the touch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher with 26 students and a new set of spring scales starts a force unit by defining net force and showing the equation, then has students explore with the scales. Which change best aligns the lesson with the 5E model?

- A. Keep it; explanation should always come first so students are not confused.
- B. Have students investigate with the scales first, then introduce net force using their observations.
- C. Begin with a graded quiz on net force.
- D. Skip exploration to save time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

In a 5E lesson, students have just explored magnets and shared their observations. Which phase comes next?

- A. Explain: build explanations and introduce terms tied to what students observed
- B. Evaluate: give the unit test
- C. Engage: spark interest with a new question
- D. Elaborate: apply the ideas to a new situation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A school received a \$40,000 grant for tablets. For a lesson on how different materials heat and cool, which use of technology best supports the lesson's science practice goal?

- A. Students type lab notes on tablets instead of paper.
- B. Use tablets for every step, since more technology always improves learning.
- C. Pick any tablet activity to justify the grant spending.
- D. Temperature probes log data in real time, and students graph and compare heating patterns.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a lab room with new epoxy countertops, students will mix dilute hydrochloric acid with magnesium ribbon. Several students wear prescription glasses. What eye protection is required?

- A. Safety glasses with side shields
- B. Prescription glasses are enough for students who wear them.
- C. None, because the acid is dilute
- D. Indirect-vent chemical splash goggles for everyone, worn over prescription glasses

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which activity requires chemical splash goggles rather than no eye protection or impact-only glasses?

- A. Heating a salt solution in a test tube
- B. Looking at leaves with a hand lens
- C. Measuring a lab table with a meter stick
- D. Only work with concentrated acid; dilute solutions never need goggles

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

During a lab in a classroom that recently got a new smartboard, a student gets a splash of dilute sodium hydroxide in one eye. What is the first action?

- A.** Read the SDS before starting to rinse.
- B.** Flush the eye at the eyewash for at least 15 minutes with eyelids held open while help is called.
- C.** Neutralize the base with a few drops of dilute acid.
- D.** Rinse for a few seconds, then return to the lab.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A new teacher inherits a stockroom built in 1985 where chemicals are shelved alphabetically, so acids sit beside bases and oxidizers beside flammables. What reorganization is correct?

- A.** Keep the alphabetical order so chemicals are easy to find.
- B.** Put all acids and bases together on one shelf because both are corrosive.
- C.** Store by compatible families: acids away from bases, oxidizers away from flammables, flammables in a rated cabinet.
- D.** Put heavy bottles on the top shelves to save space.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Why should nitric acid, a strong oxidizer, not be stored next to ethanol?

- A.** Ethanol freezes nitric acid.
- B.** Both are acids.
- C.** An oxidizer can react violently with a flammable and feed a fire.
- D.** There is no reason; any two liquids can share a shelf.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A teacher ~~whose classroom just got a new interactive projector~~ writes an objective: “Students will analyze temperature data they collected to find patterns in how sunlight heats land and water.” Which dimension is “analyze data”?

Target word: classify the verb **Decides:** It is something students do.

A	Confusion set	“Patterns” is the crosscutting concept here.
B	Adjacent category	The core idea is how sunlight heats land and water.
C	Correct	Analyzing and interpreting data is one of the eight practices.
D	Dropped element	Drops the practices dimension entirely.

2. Answer A Concept 1 • transfer

In the objective “Students will analyze temperature data they collected to find patterns in how sunlight heats land and water,” which part is the crosscutting concept?

Target word: spot the lens **Decides:** The idea that applies across all sciences.

A	Correct	Patterns is a crosscutting concept.
B	Confusion set	That is a practice.
C	Adjacent category	That is the core idea.
D	Camouflage	Equipment is not a dimension.

3. Answer B Concept 2

A team ~~at a school that runs a Saturday science fair~~ plans a unit toward a performance expectation: week 1, vocabulary for the core idea; week 2, graphing practice with unrelated data; week 3, a lecture on patterns. What is the main problem?

Target word: flaw in the unit design **Decides:** The dimensions are taught apart.

A	Inverted operation	Front-loading vocabulary deepens the problem.
B	Correct	Performance expectations call for integrated, three-dimensional learning.
C	Absolute language	How students learn the dimensions affects whether they can integrate them.
D	Wrong stem number	Length is not the core flaw.

4. Answer D Concept 3

~~At a school whose lab was renovated last summer,~~ a teacher is choosing an anchoring phenomenon for a unit on chemical reactions. Which is strongest?

Target word: best phenomenon **Decides:** Observable, puzzling, explainable with the unit’s ideas.

A	Confusion set	A topic, not a phenomenon.
B	Camouflage	Reacts to the renovation detail.
C	Adjacent category	Vocabulary, not an event to explain.
D	Correct	A real event that raises questions students can answer by the end.

5. Answer B Concept 4

A teacher ~~with 26 students and a new set of spring scales~~ starts a force unit by defining net force and showing the equation, then has students explore with the scales. Which change best aligns the lesson with the 5E model?

Target word: sequence fix **Decides:** Explore comes before Explain.

A	Absolute language	5E deliberately puts exploration first.
B	Correct	Experience first gives the terms meaning.
C	Adjacent category	Evaluate comes last, not first.
D	Dropped element	Drops a whole phase of the model.

6. Answer A Concept 4 • transfer

In a 5E lesson, students have just explored magnets and shared their observations. Which phase comes next?

Target word: next phase **Decides:** After Explore comes Explain.

A	Correct	Formal ideas follow exploration.
B	Confusion set	Evaluate is the last phase.
C	Inverted operation	Engage comes first.
D	Adjacent category	Skips the explanation phase.

7. Answer D Concept 5

A school ~~received a \$40,000 grant for tablets.~~ For a lesson on how different materials heat and cool, which use of technology best supports the lesson's science practice goal?

Target word: purposeful technology use **Decides:** Tech that lets students collect and analyze data.

A	Adjacent category	Simple substitution; learning is unchanged.
B	Absolute language	Technology helps only when it serves the goal.
C	Camouflage	Reacts to the grant rather than the goal.
D	Correct	Technology directly supports data analysis.

8. Answer D Concept 6

In a lab room ~~with new epoxy countertops,~~ students will mix dilute hydrochloric acid with magnesium ribbon. Several students wear prescription glasses. What eye protection is required?

Target word: required eye protection **Decides:** Chemicals that can splash.

A	Adjacent category	Protect against impact, not splashes.
B	Confusion set	Everyday glasses do not protect against splashes.
C	Dropped element	Drops the splash hazard; dilute acid still harms eyes.
D	Correct	Splash goggles seal around the eyes.

9. Answer A Concept 6 • transfer

Which activity requires chemical splash goggles rather than no eye protection or impact-only glasses?

Target word: when splash goggles apply **Decides:** Heated liquids can splash.

A	Correct	Heat, liquid, and glass together create a splash hazard.
B	Confusion set	Viewing equipment is not a hazard.
C	Adjacent category	No chemical or splash hazard.
D	Absolute language	Many dilute chemicals still injure eyes.

10. Answer B Concept 7

During a lab in a classroom that recently got a new smartboard, a student gets a splash of dilute sodium hydroxide in one eye. What is the first action?

Target word: first response **Decides:** Speed of flushing limits eye damage.

A	Adjacent category	The SDS matters, but flushing comes first.
B	Correct	Immediate, prolonged flushing is the standard response.
C	Inverted operation	Adds another chemical and heat to the eye.
D	Dropped element	Drops the full 15-minute flush and medical follow-up.

11. Answer C Concept 8

A new teacher inherits a stockroom built in 1985 where chemicals are shelved alphabetically, so acids sit beside bases and oxidizers beside flammables. What reorganization is correct?

Target word: storage method **Decides:** Incompatible chemicals are side by side.

A	Dropped element	Drops compatibility, the point of safe storage.
B	Confusion set	Acids and bases react with each other.
C	Correct	Separation prevents dangerous reactions if containers leak.
D	Inverted operation	Heavy and liquid containers go low.

12. Answer C Concept 8 • transfer

Why should nitric acid, a strong oxidizer, not be stored next to ethanol?

Target word: reason for separation **Decides:** Oxidizer plus flammable.

A	Adjacent category	Not a real hazard.
B	Confusion set	Ethanol is not an acid.
C	Correct	Oxidizers supply oxygen to combustion.
D	Absolute language	Compatibility rules exist because some pairs react.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The three dimensions of science learning	Science and engineering practices are what students do (analyze data, model, argue); crosscutting concepts are lenses across fields (patterns, cause and effect, scale); disciplinary core ideas are the content.	Mixing up a practice and a crosscutting concept in the same objective.	An objective with a verb, a lens word, and a content idea.
2	Performance expectations call for integrated learning	Performance expectations combine all three dimensions; lessons should have students use practices and crosscutting concepts to make sense of core ideas together.	Teaching vocabulary, skills, and concepts in separate weeks.	A unit plan that separates the dimensions.
3	Anchoring phenomena	An anchoring phenomenon is an observable, puzzling event students can explain with the unit's ideas; a topic name or vocabulary word is not one.	Using a topic title or definition as the phenomenon.	"Which is the best anchoring phenomenon?"
4	The 5E sequence	Engage, Explore, Explain, Elaborate, Evaluate: students explore before formal explanation and vocabulary.	Front-loading definitions before students explore.	A lesson that starts with a lecture, then an activity.
5	Purposeful technology integration	Technology should serve the learning goal or practice (for example, probes that let students collect and analyze real data), not novelty.	Assuming more technology or simple substitution improves learning.	New devices and a lesson goal in the same stem.
6	Eye protection	Chemical splash goggles (ANSI Z87.1, indirect vent) are required whenever chemicals, heat, or glassware are used; everyday glasses and impact-only safety glasses do not protect against splashes.	Treating prescription glasses or safety glasses as enough.	Liquids, acids, or heating in a middle school lab.
7	Emergency response to chemical exposure	For a chemical in the eye, flush immediately at the eyewash for at least 15 minutes with eyelids held open, then get medical help; consult the SDS afterward.	Neutralizing with another chemical, rinsing briefly, or reading the SDS first.	A splash to the eye or skin.
8	Chemical storage by compatibility	Store chemicals by compatible families, not alphabetically: acids apart from bases, oxidizers apart from flammables, flammables in a rated cabinet, heavy containers low.	Alphabetical shelving or grouping all corrosives together.	An inherited stockroom arranged by name.

Course scope (public description): Science Methods—Middle Grades General Science focuses on teaching methods specific to science for graduate students seeking an endorsement in middle school science. Course content focuses on the design and teaching of standards- based lessons using the three dimensions of science (science and engineering practices, crosscutting concepts, and disciplinary core ideas) and the appropriate integration of technology into those lessons. Students in this course work within their content areas to evaluate, enhance, and plan appropriate science instruction. This course includes laboratory safety training and certification, which includes safe laboratory practices and procedures for science classrooms and the proper use of personal protective equipment. Previous coursework in curriculum, instruction, and assessment is a prerequisite for this course.